

Abstract No.

Systematic DNA Methylation Profiling Improves Diagnostic Precision in Paediatric Low-Grade Glioma: Results from the HIT-LOGGIC Registry

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Background:

Paediatric low-grade gliomas (pLGG) represent the most common CNS tumours in children. Despite established histopathological diagnostics, diagnostic uncertainty persists, particularly for rare entities. The added value of systematic DNA methylation profiling in a population-based diagnostic setting has not been comprehensively quantified.

Methods:

We analysed 1,173 tumours from the prospective, near population-based, Germany-wide HIT-



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LOGGIC registry over a five-year period, integrating histopathology with systematic DNA methylation profiling. We assessed the impact of methylation profiling on diagnostic precision, CNS WHO grading, and epidemiological classification.

Results:

DNA methylation profiling changed the diagnosis in 5.7% of externally diagnosed cases, including 22 patients with a change in CNS WHO grade, and enabled definitive classification in 71.7% of 205 histologically ambiguous tumours. Five high-grade tumours (0.4%) were newly identified that had not been recognised as such by histology alone. Integration of methylation profiling substantially expanded and diversified the diagnostic spectrum, with marked increases in the detection of rare entities such as diffuse leptomeningeal glioneuronal tumour (DLGNT), pleomorphic xanthoastrocytoma (PXA), and diffuse astrocytoma with MYB/MYBL1 alteration (LGG MYB), refining epidemiological estimates for paediatric low-grade CNS tumours.

Conclusion:

Systematic implementation of DNA methylation profiling in paediatric neuro-oncology improves diagnostic precision, supports clinical decision-making, and refines epidemiological estimates in pLGG, supporting its adoption as a routine component of diagnostic workflows.